

KCPC Week 1 Reading

Introduction to Competitive Programming, Invariants, and Logic

This document contains curated supplementary readings for Workshop 1. These texts expand upon the core themes introduced in the session: competitive programming fundamentals, invariant reasoning, epistemic logic (common knowledge), and impartial combinatorial games.

I. Competitive Programming Fundamentals

Competitive Programmer's Handbook (Antti Laaksonen)

<https://cses.fi/book/book.pdf>

- **Chapter 1: Introduction** Contest environment, basic language features, and input/output optimisation.
- **Chapter 2: Time Complexity** Asymptotic notation (O -notation), estimating operation counts per second (10^8 ops/sec rule), and detecting TLE bounds.

Competitive Programming 4 (Steven Halim, Felix Halim, Suhendry)

<https://cpbook.net/>

- **Chapter 1: Introduction** Contest categories (ICPC, IOI, Codeforces, AtCoder), standard submission verdicts, and typical problem pitfalls.

II. Invariants and Mathematical Heuristics

Problem-Solving Strategies (Arthur Engel)

<https://link.springer.com/book/10.1007/978-0-387-21607-2>

- **Chapter 1: The Invariance Principle** Foundational treatment of invariants, monovariants, and coloring arguments in discrete dynamical processes.

The Art and Craft of Problem Solving (Paul Zeitz)

John Wiley & Sons, 2006.

- **Chapter 3: Tactics for Solving Problems** Symmetry, parity, extremal thinking, and simplifying assumptions.

III. Epistemic Logic and Common Knowledge

Reasoning About Knowledge (Fagin, Halpern, Moses, Vardi)

MIT Press, 2003. <https://mitpress.mit.edu/9780262562003/reasoning-about-knowledge/>

- **Chapter 1: Introduction** The distinction between mutual knowledge and common knowledge; the formal mud-on-the-forehead puzzle.
- **Chapter 2: A Model for Knowledge** Kripke structures and epistemic possible worlds.

IV. Combinatorial Game Theory

Game Theory (Thomas S. Ferguson)

https://www.math.ucla.edu/~tom/Game_Theory/comb.pdf

- **Part I: Impartial Combinatorial Games** Subtraction games, normal-play conventions, N - and P -positions, and backward induction on directed acyclic graphs.

CP-Algorithms: Games on Arbitrary Graphs

https://cp-algorithms.com/game_theory/games_on_graphs.html

- Topological determination of winning and losing states via retro-analysis / backward induction.

